

Student: _____ Student ID#: _____

EXAM 2

README

1. The answers to the questions in this exam must be your own. **Violations will result in a Fail grade in the course** and will be handled according to NAU's Academic Integrity Policy.
2. Sign this first page with your name and NAU id **before the test starts**.
3. Read the instructions and questions carefully. Correctly interpreting the questions is part of the examination. **I will NOT answer questions during the exam.**
4. The exam starts at 10:00 AM and ends at 12:00 PM. If you write in your exam after time runs up, your exam may be taken from you, and you'll receive a ZERO as a grade.
5. Phones, laptops, or any other electronic devices are not allowed during exam time. Please turn your phones off and store them out of your reach. If you use any devices during exam time, the exam may be taken from you, and you'll receive a zero as a grade.
6. Provide your answer using a non-erasable pen. If you use pencil or erasable pen, you cannot request grade review.
7. The exam is **individual** and closed book, closed notes.
8. Your grade will appear on BBLearn as soon as all the tests are graded. Don't email me asking for grades, if it's not on BBLearn, I don't have them just yet. Thanks for your patience.
9. There is **only one correct answer** for each question.
10. **No partial marks** are possible in the questions.

CS 599 – Contemporary developments
Principles of Programming Languages

1. (4 points) About the basics of programming languages

- I. Are designed only to enable the communication among developers
- II. Programming languages are less expressive than natural language
- III. In the beginning the programming languages were closer to machine-level commands and later they evolved to enable developers' readability

- (a) Only I is correct
- (b) Only I and II are correct
- (c) All the items are correct
- (d) Only items I and III are correct
- (e) II and III are correct**

2. (4 points) About the evaluation criteria for choosing a programming language, mark the correct sentence:

- (a) Reliability of a programming language is closely related to the level of support that the community offers
- (b) To make a programming language simpler it is necessary to make it closer to natural language, increasing the number of basic constructs
- (c) Too much orthogonality may result in languages with lower writability**
- (d) Support is not an important criterion to choose a programming language
- (e) Since programming languages are usually open source, cost is not a problem when choosing a programming language

3. (8 points) Match the columns add the corresponding letter on the left to the right column

A. Interpreted Languages

B. Compiled Languages

B	The execution is, in general, faster
A	The code is generated dynamically and can be executed piece by piece
B	A binary code is generated and the program is executed from this binary
A	Debug the source code at run time, line by line
A	The program needs to be executed from the source code every time
B	The code generated is specific hardware architecture
B	C is an example of this type of languages
A	PHP and Python are examples of this type of language

4. (4 points) About the historical perspective of Programming Languages, check correct sentence

- (a) Machine language-based languages were the first ones, usually having commands that were easier for developers understand
- (b) The von Neumann architecture was not considered when programming languages were created
- (c) COBOL was created to deal with the scientific problems at the 2000's
- (d) High-level languages were introduced to make the maintenance easier**
- (e) All the languages we use currently are from the 50's, 60's, or 70's, no new languages had been created after that.

CS 599 – Contemporary developments
Principles of Programming Languages

5. (8 points) Consider the following grammar:

$\langle S \rangle \rightarrow a\langle A \rangle\langle B \rangle b$

$\langle A \rangle \rightarrow \langle A \rangle b \mid b$

$\langle B \rangle \rightarrow a\langle B \rangle \mid a$

Are these sentences part of the language defined by the grammar? Show the leftmost derivation for each of them (step-by-step, changing one symbol per derivation) and conclude if they are part of the language or not. **Answers with no derivation will receive ZERO.**

abaab $\langle S \rangle$ a $\langle A \rangle$ $\langle B \rangle$ b a b $\langle B \rangle$ b a b a $\langle B \rangle$ b a b a a b <i>The sentence is part of then language</i>	abbbabb $\langle S \rangle$ a $\langle A \rangle$ $\langle B \rangle$ b a $\langle A \rangle$ b $\langle B \rangle$ b a $\langle A \rangle$ b b $\langle B \rangle$ b a b b b $\langle B \rangle$ b a b b b a $\langle B \rangle$ b <i>Impossible to get a b after the “a” since $\langle B \rangle$ derivations let to a’s only</i>
---	---

6. (4 points) Consider the following sentences about carbon footprint/green IT and programming languages:

- I. The carbon footprint is influenced by the tools the hardware used and programming languages do not have influence on that
- II. Python is not very efficient, consuming way more energy than C, for example
- III. The resources (memory, computation time) used by the programming language influence on the energy consumption and, therefore, the carbon footprint

we can say that:

- (a) Only item I is correct
- (b) Items I and II are correct, and III is incorrect
- (c) Items I and III are correct, II is not
- (d) All items are correct
- (e) **Items II and III are correct, but item I is not**

CS 599 – Contemporary developments
Principles of Programming Languages

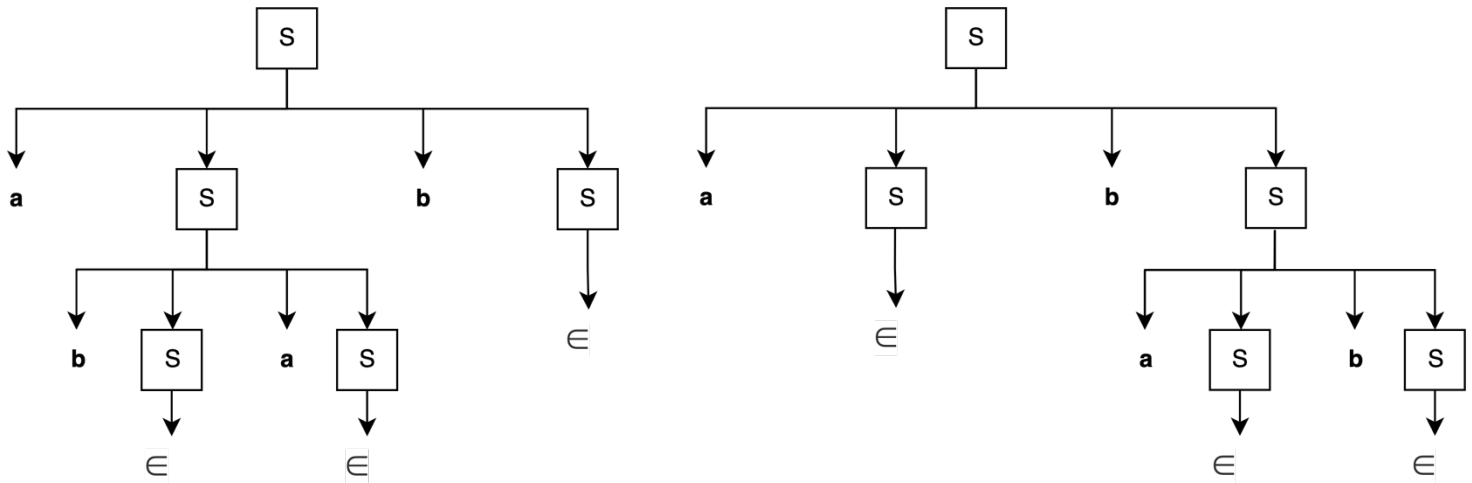
7. (8 points) Prove that the following grammar is ambiguous

$S \rightarrow aSbS$

$S \rightarrow bSaS$

$S \rightarrow \epsilon$

It is possible to derive 2 different trees for the word **abab**:



CS 599 – Contemporary developments
Principles of Programming Languages

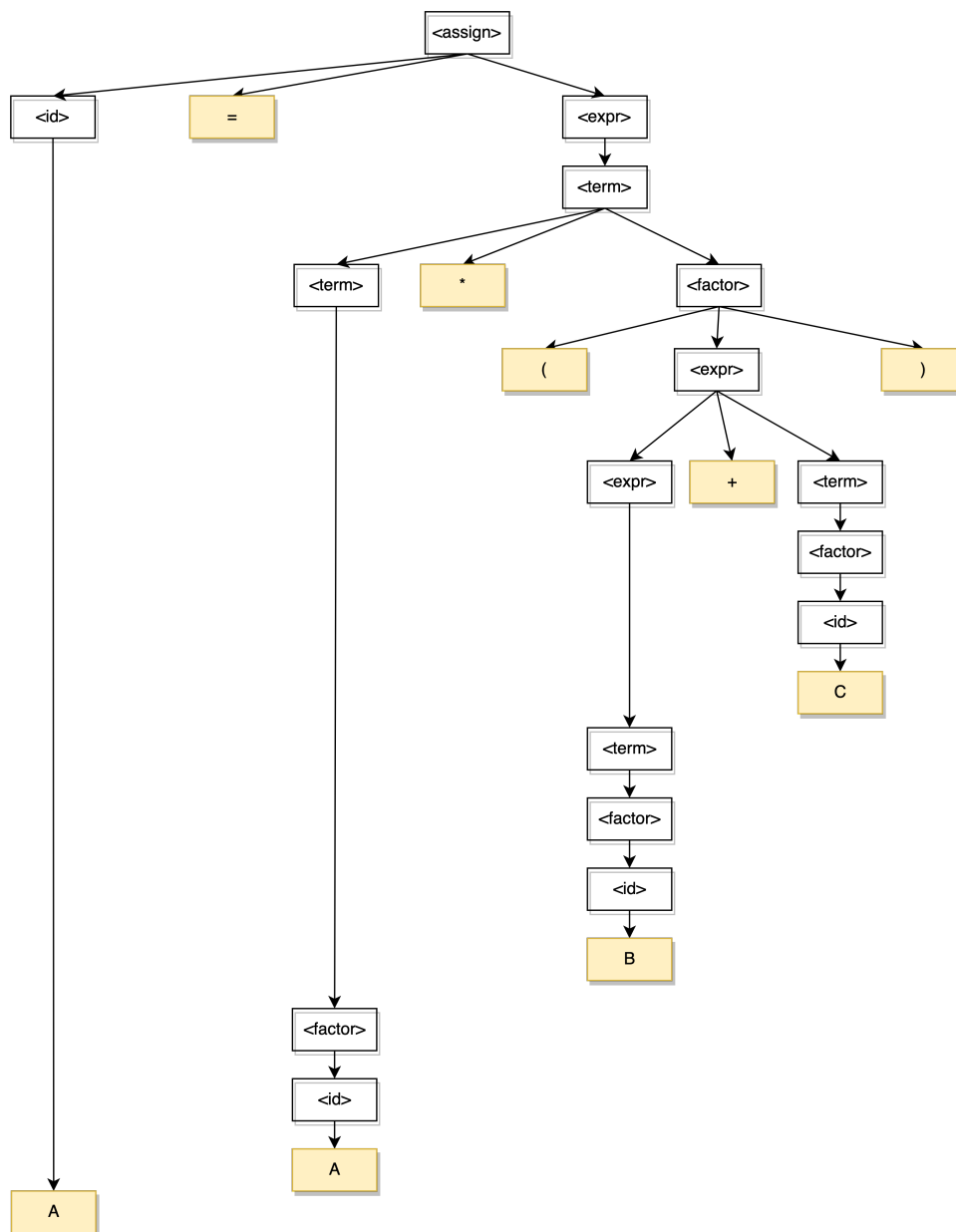
8. (8 points) Given the following grammar

```

<assign>  →  <id> = <expr>
<id>      →  A |
              B |
              C
<expr>    →  <expr> + <term> |
              <term>
<term>    →  <term> * <factor> |
              <factor>
<factor>  →  ( <expr> ) |
              <id>
    
```

Create the parse tree (rightmost derivation) considering for the grammar and sentence below:

A = A * (B + C)



CS 599 – Contemporary developments
Principles of Programming Languages

9. (4 points) About imperative programming, it is **incorrect** to say that:

- (a) It follows a process of describing steps that simply change the state of the computer.
- (b) It is generally a process of developing software simply by composing functions.**
- (c) It mainly focuses on describing how the program executes step-by-step.
- (d) These programs are structured as successive assignments of values to variable names.
- (e) It uses statements that change a program's state.

10. (8 points) Match the paradigm on the left with the sentences on the right

A. Imperative Programming	B	Generally structured as successive nested function calls.
B. Functional Programming	A	Based on a paradigm where each statement is sequentially evaluated
C. Logic Programming	B	Repetition happens only by recursion not by loops sequencing commands
	B	Referential Transparency is one of its characteristics
	A	Based on the Von Neumann architecture model
	C	Relies on facts and relations between objects
	C	Prolog is the representative of this paradigm
	A	Uses statements that change a program's state.

11. (4 points) Consider the following sentences about inheritance

- I. A subclass inherits variables and functions from a superclass
- II. A private function from a superclass is not inherited by the subclass
- III. Multiple inheritance is only possible in theory, since it is not possible to solve the diamond problem
- IV. One way of implementing polymorphism is by overriding a function from a superclass.

we can say that:

- (a) Only I and II are correct
- (b) I, II and III are correct, but IV is incorrect
- (c) I, II and IV are correct, but III is not**
- (d) II and IV are the only correct sentences
- (e) All the sentences are correct

CS 599 – Contemporary developments
Principles of Programming Languages

12. (4 points) About object-oriented programming, check the incorrect alternative

- (a) The object-orientated paradigm is based on the concept of object, which encapsulates both data and the functions to manipulate its data
- (b) In Object Oriented programs it is possible to determine the visibility of a variables and function, varying from public (visible to any other object) to private (visible only inside of the object).
- (c) Object orientation is one type of declarative programming languages**
- (d) Class is a model for a set of objects
- (e) Inheritance allows a class to reuse and extend the variables and functions of another class

13. (8 points) Given the following code in Haskell:

```
func :: Integer -> Integer
func n
  | n == 0 = 0
  | n == 1 = 1
  | otherwise = func (n-1) + func (n-2)
```

a. What is the output when we call `func 4`

3 *(This is an adaptation of the Fibonacci function presented in class)*

b. What does `Integer -> Integer` in the first line represent?

The domain and the range

OR: type of the input and type of return

14. (4 points) Given the following facts and rule in Prolog:

parent(a,b) . parent(c,b) . parent(b,d) . parent(b,e) . parent(d,f) .	grandchild(X,Y) :- parent(Y,Z) , parent(Z,X) .
---	--

What would be the result of the query:
`grandchild(G,a)`

d
e
d and e are the grandchild of a (a is parent of b, b is parent of d and e)

CS 599 – Contemporary developments
Principles of Programming Languages

15. (4 points) Given the following code in Prolog:

```
func(1, 1).  
func(N, Result) :-  
    NEXT is N-1,  
    func(NEXT, Return),  
    Result is N * Return.
```

What is the output when we call `func(4, R)`

24 (*This is a factorial function that I presented in class*)

16. (4 points) About type binding and variable declaration:

- I. One advantage of implicit declaration is that you can change the type of the variable at any point of the program
- II. Explicit declaration means that a program statement is used for declaring/defining the types of variables
- III. In static type binding a variable receives a type when it is first declared, and the type never changes
- IV. Python used dynamic type binding and explicit declaration

we can say that:

- (a) Only I and II are correct
- (b) I, II and III are correct, but IV is incorrect
- (c) II, III and IV are correct, but I is not**
- (d) I and III are the only correct sentences
- (e) Only II and III are correct

CS 599 – Contemporary developments
Principles of Programming Languages

17. (4 points) Considering that this Python program is executed using **dynamic scoping**, what would be the output?

```
a = 5
b = 4

def f_a():
    a = 7
    print (a + b)
    c = f_b()
    return c

def f_b():
    b = 2
    return f_c()

def f_c():
    return a + b

print (f_a())
```

11

9

18. (8 points) Mark True (T) or False (F) for the following sentences about lexical and syntax analysis

- [F] The output of Lexical Analysis step a Parse Tree and entries in the Symbol Table
- [F] The lexical analyzer breaks sentences into lexemes (words) and identify the meaning of a complete sentence
- [T] To identify a token, the lexical analyzer will read character per character until it finds the end of a lexeme (usually a whitespace)
- [F] Comments are defined as part of the grammar and are analyzed by the lexical analyzer
- [T] After the lexical analyzer finishes its job, the syntax analysis takes place in which the tokens are analyzed giving an abstract syntax tree as output
- [F] The syntax analyzer checks the tokens individually, not validating if the sequence in which they appear is correct
- [T] A syntax parse tree is built according to the grammar, considering the lexical units
- [T] An Abstract Syntax Tree (AST) is smaller than the Parse Tree because it removes the "syntactic sugar" from the tree